POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 3021 | Semester III | Theory | 4 Credits

Thermal Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 3021 as **Thermal Engineering** in Semester III for Mechanical Engineering. At preparation time, the official SITTTR programme page could not be retrieved because its server repeatedly terminated the TLS connection. With user approval, handbook content uses an AICTE-supported diploma Thermal Engineering text and a public diploma curriculum; it is not represented as recovered official SITTTR Course 3021 detailed-syllabus content.

Source treatment: Because official Course 3021 teaching-scheme and detailed-content data were temporarily inaccessible, contact hours, credits, CIE/ESE and teaching scheme are provisional handbook placeholders aligned with neighbouring Revision 2026 third-semester theory courses. Module coverage, explanations and activities come from the cited public diploma Thermal Engineering sources and must be reconciled with official SITTTR material when it becomes accessible.

Verified source note: The SITTTR Revision 2026 programme scheme was accessible earlier for Civil Engineering, but the Mechanical Engineering programme request failed during the Course 3021 cycle due to repeated TLS server termination. This limitation and the approved supplementary sources are stated here for traceability.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Thermal Engineering connects thermodynamic principles with practical energy-conversion equipment. It develops understanding of energy systems, ideal gases and steam, boilers and turbines, heat transfer, heat exchangers and internal-combustion engines.

> **Malayalam support:** ▯ handbook ▯▯▯▯▯▯▯▯▯▯▯▯ syllabus topic ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯▯▯▯▯▯ principle, diagram/procedure, application, common mistake ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯.

Prerequisite foundation

Engineering mathematics, physics, basic mechanics and clear use of SI units.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain thermodynamic systems, properties and laws in engineering contexts.
2. Apply ideal-gas and steam concepts using diagrams, tables and basic calculations.
3. Describe steam-power-plant equipment and its safe, efficient operation.
4. Explain heat transfer, heat exchangers and internal-combustion-engine principles.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain thermodynamic systems, properties and laws and relate them to energy devices.	Understand
CO2	Apply ideal-gas-process and steam-property concepts in elementary thermal calculations.	Apply
CO3	Describe steam power-plant components, boilers, turbines, condensers and cooling towers.	Understand
CO4	Explain heat transfer and select heat-exchanger arrangements for basic applications.	Apply
CO5	Explain power cycles and construction/working of SI and CI internal-combustion engines.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–
CO5	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Thermodynamic Fundamentals	Systems, properties, state/process/cycle, heat/work, enthalpy/entropy and laws of thermodynamics.	Approx. 12 hours	CO1	Module 1
2	Ideal Gases and Steam Fundamentals	Gas equation and processes; P–V/T–S diagrams; steam formation, types/properties, steam tables and Rankine-cycle overview.	Approx. 14 hours	CO2	Module 2
3	Steam Power Plants and Heat Transfer	Plant layout; boilers, nozzles, turbines, condensers, cooling towers; conduction, convection, radiation and heat exchangers.	Approx. 18 hours	CO3	Module 3
4	Power Cycles and Internal-Combustion Engines	Carnot/Otto/Diesel/Dual cycles; engine terminology; SI/CI classification; two-stroke/four-stroke construction and working.	Approx. 16 hours	CO5	Module 4

LEARNING ROADMAP

How the Modules Connect

Thermodynamic Fundamentals

Systems, properties, state/process/cycle, heat/work, enthalpy/entropy and laws of thermodynamics.

CO1 · Approx. 12 hours

Ideal Gases and Steam Fundamentals

Gas equation and processes; P–V/T–S diagrams; steam formation, types/properties, steam tables and Rankine-cycle overview.

CO2 · Approx. 14 hours

Steam Power Plants and Heat Transfer

Plant layout; boilers, nozzles, turbines, condensers, cooling towers; conduction, convection, radiation and heat exchangers.

CO3 · Approx. 18 hours

Power Cycles and Internal-Combustion Engines

Carnot/Otto/Diesel/Dual cycles; engine terminology; SI/CI classification; two-stroke/four-stroke construction and working.

CO5 · Approx. 16 hours

MODULE 1

Thermodynamic Fundamentals

Systems, properties, state/process/cycle, heat/work, enthalpy/entropy and laws of thermodynamics.

Approx. 12 hours **CO1** **Understand · Apply**

Learning goals

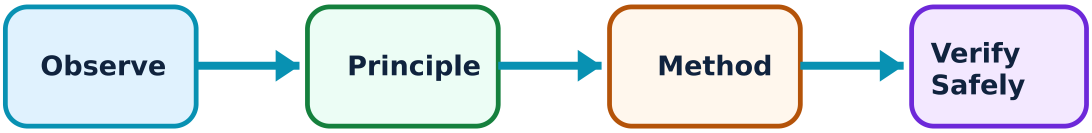
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Thermodynamic Fundamentals: identify the system, state its principle, follow the correct method and verify the result safely.

1. Systems, properties, process and cycle−

A thermodynamic system is a chosen quantity of matter or region in space. Open, closed and isolated systems differ by mass/energy interaction. State, property, process and cycle provide the vocabulary for analysing thermal equipment.

Study and application method

Draw a system boundary, classify the system, list relevant intensive/extensive properties and identify inputs/outputs before writing an energy relation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a system boundary, classify the system, list relevant intensive/extensive properties and identify inputs/outputs before writing an energy relation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Do not confuse a property (state quantity) with a path-dependent transfer such as heat or work.

2. Heat, work, enthalpy and entropy−

Heat and work are energy-transfer modes. Enthalpy is useful for flowing systems; entropy helps express directionality and irreversibility. Correct sign convention and system definition are essential.

Study and application method

State sign convention, label transfer across system boundary and use units consistently. For a flow device, list inlet/outlet states and work/heat terms before simplification.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Heat is not stored ‘inside’ a body as a property; internal energy and enthalpy are state properties.

3. Laws of thermodynamics and energy devices−

The zeroth law supports temperature measurement; the first law expresses energy conservation; the second law constrains heat-engine, refrigerator and heat-pump operation. These laws guide analysis of boilers, turbines and condensers.

Study and application method

Name the governing law, state it in appropriate words/equation and relate it to one device with inputs, outputs and expected energy conversion.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Name the governing law, state it in appropriate words/equation and relate it to one device with inputs, outputs and expected energy conversion.

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Common mistake / precaution: A thermal device cannot convert all supplied heat into useful work in a cycle; do not ignore second-law limitation.

Mechanical-work calculator

Force (N)

10

Displacement (m)

2

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Ideal Gases and Steam Fundamentals

Gas equation and processes; P–V/T–S diagrams; steam formation, types/properties, steam tables and Rankine-cycle overview.

Approx. 14 hours | CO2 | Understand · Apply

Learning goals

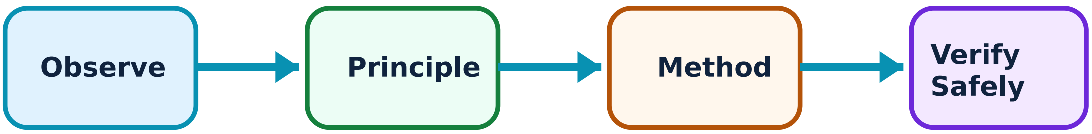
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Ideal Gases and Steam Fundamentals: identify the system, state its principle, follow the correct method and verify the result safely.

1. Ideal-gas equation and processes−

An ideal gas is modelled using the relation between pressure, volume and temperature. Isobaric, isochoric, isothermal, isentropic and polytropic processes are represented on property diagrams and may involve heat/work changes.

Study and application method

List initial/final state data, identify process constraint, choose the corresponding relation and show every unit conversion before calculating work or energy change.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List initial/final state data, identify process constraint, choose the corresponding relation and show every unit conversion before calculating work or energy change.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not apply the ideal-gas model indiscriminately near phase change or when data require real-fluid properties.

2. Steam formation and properties−

At constant pressure, heating water produces sensible heating, phase change and potentially superheated steam. Wet, dry saturated and superheated steam have different property descriptions; dryness fraction characterises wet steam.

Study and application method

Locate state conceptually on a temperature–enthalpy or temperature–entropy diagram, identify steam type and use a steam table carefully with correct pressure/temperature entry.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Locate state conceptually on a temperature–enthalpy or temperature–entropy diagram, identify steam type and use a steam table carefully with correct pressure/temperature entry.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not use a saturation-table value for superheated steam without checking the state region.

3. Rankine-cycle overview−

The Rankine cycle is a basic model for steam-power conversion involving boiler, turbine, condenser and pump. Diagrams help connect energy addition, expansion, rejection and pumping.

Study and application method

Draw the four main components in flow sequence, label state points and describe energy transfer at each component before interpreting the diagram.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw the four main components in flow sequence, label state points and describe energy transfer at each component before interpreting the diagram.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A schematic cycle is not a detailed plant-design calculation; state assumptions clearly.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Steam Power Plants and Heat Transfer

Plant layout; boilers, nozzles, turbines, condensers, cooling towers; conduction, convection, radiation and heat exchangers.

Approx. 18 hours | CO3 | Understand · Apply

Learning goals

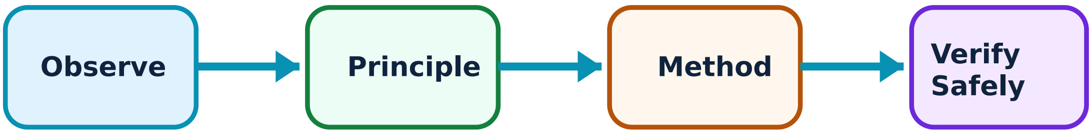
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Steam Power Plants and Heat Transfer: identify the system, state its principle, follow the correct method and verify the result safely.

1. Steam power-plant layout and boilers−

A steam power plant converts chemical/thermal energy into electrical output through coordinated boiler, turbine, condenser and auxiliary systems. Boilers generate steam under controlled conditions; mountings and accessories support safety and performance.

Study and application method

Draw a block layout showing flow of fuel/air, water/steam and cooling medium; then describe one boiler's working path and one safety consideration.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a block layout showing flow of fuel/air, water/steam and cooling medium; then describe one boiler's working path and one safety consideration.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Boiler operation is safety-critical; do not treat construction diagrams as operating instructions.

2. Nozzles, turbines, condensers and cooling towers−

Nozzles accelerate steam, turbines extract shaft work, condensers reject heat and maintain low exhaust pressure, and cooling towers reject heat to air. Their functions are connected in the power-plant energy path.

Study and application method

Use function → construction feature → working → application structure for each device. Compare impulse/reaction or condenser/cooling-tower roles with a table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use function → construction feature → working → application structure for each device. Compare impulse/reaction or condenser/cooling-tower roles with a table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse a condenser, which exchanges heat/condenses vapour, with a cooling tower, which rejects heat from cooling water.

3. Modes of heat transfer and heat exchangers−

Conduction transfers heat through material; convection involves fluid motion; radiation transfers energy by electromagnetic emission. Heat exchangers transfer heat between fluids without intended mixing in typical arrangements.

Study and application method

Identify dominant mode and boundary conditions, draw a simple thermal-resistance path for conduction, and compare shell-and-tube versus plate heat exchanger by flow path and application.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify dominant mode and boundary conditions, draw a simple thermal-resistance path for conduction, and compare shell-and-tube versus plate heat exchanger by flow path and application.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Use temperature difference direction carefully; heat flows spontaneously from higher to lower temperature.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Power Cycles and Internal-Combustion Engines

Carnot/Otto/Diesel/Dual cycles; engine terminology; SI/CI classification; two-stroke/four-stroke construction and working.

Approx. 16 hours **CO5** **Understand · Apply**

Learning goals

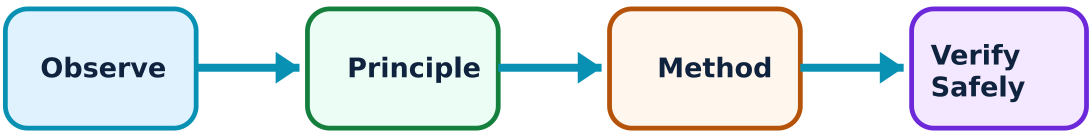
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Power Cycles and Internal-Combustion Engines: identify the system, state its principle, follow the correct method and verify the result safely.

1. Air-standard power cycles−

Idealised Carnot, Otto, Diesel and Dual cycles represent energy conversion in heat engines. P–V and T–S diagrams show compression, heat addition, expansion and rejection in simplified form.

Study and application method

Draw axes and state sequence of processes, label high/low-temperature regions and compare the cycle’s idealised heat-addition assumption.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw axes and state sequence of processes, label high/low-temperature regions and compare the cycle’s idealised heat-addition assumption.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not present ideal-cycle diagrams as exact representation of a real engine’s valve/combustion events.

2. IC-engine classification and terminology−

Internal-combustion engines can be classified by ignition method, stroke cycle, fuel, cooling, cylinder arrangement and application. Bore, stroke, clearance volume, compression ratio and dead centres describe engine geometry.

Study and application method

Build a classification tree and label a piston-cylinder sketch. Define each term beside its physical location rather than memorising words alone.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

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Common mistake / precaution: Spark ignition and compression ignition describe ignition principle, not merely fuel name.

3. Two-stroke and four-stroke SI/CI engine working−

Four-stroke engines separate intake, compression, power and exhaust over two crank revolutions; two-stroke engines complete a cycle more frequently with port/flow-control differences. SI and CI engines differ in mixture preparation and ignition.

Study and application method

Use one labelled sequence diagram per engine type, state piston movement/valve or port status and explain when combustion/ignition occurs.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use one labelled sequence diagram per engine type, state piston movement/valve or port status and explain when combustion/ignition occurs.

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Common mistake / precaution: Avoid mixing valve timing, port timing and stroke sequence from different engine types.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Thermodynamic Fundamentals

Use the concept flow to revise observation, principle, method and verification.

Module 2: Ideal Gases and Steam Fundamentals

Use the concept flow to revise observation, principle, method and verification.

Module 3: Steam Power Plants and Heat Transfer

Use the concept flow to revise observation, principle, method and verification.

Module 4: Power Cycles and Internal-Combustion Engines

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Create a system-boundary chart for boiler, turbine, condenser, compressor and refrigerator, identifying energy interactions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Plot/interpret simple ideal-gas process diagrams and practise steam-table state identification.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a labelled steam-power-plant flow diagram and compare boiler, turbine, condenser and cooling-tower functions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Make a two-stroke/four-stroke SI/CI comparison chart including cycle sequence and a safety/environment note.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Thermodynamic Fundamentals

Explain Systems, properties, process and cycle and state one correct method, application or precaution.−

A thermodynamic system is a chosen quantity of matter or region in space. Open, closed and isolated systems differ by mass/energy interaction. State, property, process and cycle provide the vocabulary for analysing thermal equipment.

Answer framework: Draw a system boundary, classify the system, list relevant intensive/extensive properties and identify inputs/outputs before writing an energy relation.

Important point: Do not confuse a property (state quantity) with a path-dependent transfer such as heat or work.

Explain Heat, work, enthalpy and entropy and state one correct method, application or precaution.−

Heat and work are energy-transfer modes. Enthalpy is useful for flowing systems; entropy helps express directionality and irreversibility. Correct sign convention and system definition are essential.

Answer framework: State sign convention, label transfer across system boundary and use units consistently. For a flow device, list inlet/outlet states and work/heat terms before simplification.

Important point: Heat is not stored ‘inside’ a body as a property; internal energy and enthalpy are state properties.

Explain Laws of thermodynamics and energy devices and state one correct method, application or precaution.−

The zeroth law supports temperature measurement; the first law expresses energy conservation; the second law constrains heat-engine, refrigerator and heat-pump operation. These laws guide analysis of boilers, turbines and condensers.

Answer framework: Name the governing law, state it in appropriate words/equation and relate it to one device with inputs, outputs and expected energy conversion.

Important point: A thermal device cannot convert all supplied heat into useful work in a cycle; do not ignore second-law limitation.

Module 2 — Ideal Gases and Steam Fundamentals

Explain Ideal-gas equation and processes and state one correct method, application or precaution.−

An ideal gas is modelled using the relation between pressure, volume and temperature. Isobaric, isochoric, isothermal, isentropic and polytropic processes are represented on property diagrams and may involve heat/work changes.

Answer framework: List initial/final state data, identify process constraint, choose the corresponding relation and show every unit conversion before calculating work or energy change.

Important point: Do not apply the ideal-gas model indiscriminately near phase change or when data require real-fluid properties.

Explain Steam formation and properties and state one correct method, application or precaution.−

At constant pressure, heating water produces sensible heating, phase change and potentially superheated steam. Wet, dry saturated and superheated steam have different property descriptions; dryness fraction characterises wet steam.

Answer framework: Locate state conceptually on a temperature–enthalpy or temperature–entropy diagram, identify steam type and use a steam table carefully with correct pressure/temperature entry.

Important point: Do not use a saturation-table value for superheated steam without checking the state region.

Explain Rankine-cycle overview and state one correct method, application or precaution.−

The Rankine cycle is a basic model for steam-power conversion involving boiler, turbine, condenser and pump. Diagrams help connect energy addition, expansion, rejection and pumping.

Answer framework: Draw the four main components in flow sequence, label state points and describe energy transfer at each component before interpreting the diagram.

Important point: A schematic cycle is not a detailed plant-design calculation; state assumptions clearly.

Module 3 — Steam Power Plants and Heat Transfer

Explain Steam power-plant layout and boilers and state one correct method, application or precaution.−

A steam power plant converts chemical/thermal energy into electrical output through coordinated boiler, turbine, condenser and auxiliary systems. Boilers generate steam under controlled conditions; mountings and accessories support safety and performance.

Answer framework: Draw a block layout showing flow of fuel/air, water/steam and cooling medium; then describe one boiler's working path and one safety consideration.

Important point: Boiler operation is safety-critical; do not treat construction diagrams as operating instructions.

Explain Nozzles, turbines, condensers and cooling towers and state one correct method, application or precaution.−

Nozzles accelerate steam, turbines extract shaft work, condensers reject heat and maintain low exhaust pressure, and cooling towers reject heat to air. Their functions are connected in the power-plant energy path.

Answer framework: Use function → construction feature → working → application structure for each device. Compare impulse/reaction or condenser/cooling-tower roles with a table.

Important point: Do not confuse a condenser, which exchanges heat/condenses vapour, with a cooling tower, which rejects heat from cooling water.

Explain Modes of heat transfer and heat exchangers and state one correct method, application or precaution.−

Conduction transfers heat through material; convection involves fluid motion; radiation transfers energy by electromagnetic emission. Heat exchangers transfer heat between fluids without intended mixing in typical arrangements.

Answer framework: Identify dominant mode and boundary conditions, draw a simple thermal-resistance path for conduction, and compare shell-and-tube versus plate heat exchanger by flow path and application.

Important point: Use temperature difference direction carefully; heat flows spontaneously from higher to lower temperature.

Module 4 — Power Cycles and Internal-Combustion Engines

Explain Air-standard power cycles and state one correct method, application or precaution.−

Idealised Carnot, Otto, Diesel and Dual cycles represent energy conversion in heat engines. P–V and T–S diagrams show compression, heat addition, expansion and rejection in simplified form.

Answer framework: Draw axes and state sequence of processes, label high/low-temperature regions and compare the cycle's idealised heat-addition assumption.

Important point: Do not present ideal-cycle diagrams as exact representation of a real engine's valve/combustion events.

Explain IC-engine classification and terminology and state one correct method, application or precaution.−

Internal-combustion engines can be classified by ignition method, stroke cycle, fuel, cooling, cylinder arrangement and application. Bore, stroke, clearance volume, compression ratio and dead centres describe engine geometry.

Answer framework: Build a classification tree and label a piston-cylinder sketch. Define each term beside its physical location rather than memorising words alone.

Important point: Spark ignition and compression ignition describe ignition principle, not merely fuel name.

Explain Two-stroke and four-stroke SI/CI engine working and state one correct method, application or precaution.−

Four-stroke engines separate intake, compression, power and exhaust over two crank revolutions; two-stroke engines complete a cycle more frequently with port/flow-control differences. SI and CI engines differ in mixture preparation and ignition.

Answer framework: Use one labelled sequence diagram per engine type, state piston movement/valve or port status and explain when combustion/ignition occurs.

Important point: Avoid mixing valve timing, port timing and stroke sequence from different engine types.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Thermodynamic Fundamentals.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Ideal Gases and Steam Fundamentals.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Steam Power Plants and Heat Transfer.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Power Cycles and Internal-Combustion Engines.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Systems, properties, state/process/cycle, heat/work, enthalpy/entropy and laws of thermodynamics..
2. Develop a complete answer or practical plan for: Gas equation and processes; P–V/T–S diagrams; steam formation, types/properties, steam tables and Rankine-cycle overview..
3. Develop a complete answer or practical plan for: Plant layout; boilers, nozzles, turbines, condensers, cooling towers; conduction, convection, radiation and heat exchangers..
4. Develop a complete answer or practical plan for: Carnot/Otto/Diesel/Dual cycles; engine terminology; SI/CI classification; two-stroke/four-stroke construction and working..

LAST-MILE REVISION

Quick Revision

Module 1: Thermodynamic Fundamentals

Systems, properties, state/process/cycle, heat/work, enthalpy/entropy and laws of thermodynamics.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Ideal Gases and Steam Fundamentals

Gas equation and processes; P–V/T–S diagrams; steam formation, types/properties, steam tables and Rankine-cycle overview.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Steam Power Plants and Heat Transfer

Plant layout; boilers, nozzles, turbines, condensers, cooling towers; conduction, convection, radiation and heat exchangers.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Power Cycles and Internal-Combustion Engines

Carnot/Otto/Diesel/Dual cycles; engine terminology; SI/CI classification; two-stroke/four-stroke construction and working.

- Match each topic to CO5.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Systems, properties, process and cycle	A thermodynamic system is a chosen quantity of matter or region in space.
Heat, work, enthalpy and entropy	Heat and work are energy-transfer modes.
Laws of thermodynamics and energy devices	The zeroth law supports temperature measurement; the first law expresses energy conservation; the second law constrains heat-engine, refrigerator and heat-pump operation.
Ideal-gas equation and processes	An ideal gas is modelled using the relation between pressure, volume and temperature.
Steam formation and properties	At constant pressure, heating water produces sensible heating, phase change and potentially superheated steam.
Rankine-cycle overview	The Rankine cycle is a basic model for steam-power conversion involving boiler, turbine, condenser and pump.
Steam power-plant layout and boilers	A steam power plant converts chemical/thermal energy into electrical output through coordinated boiler, turbine, condenser and auxiliary systems.
Nozzles, turbines, condensers and cooling towers	Nozzles accelerate steam, turbines extract shaft work, condensers reject heat and maintain low exhaust pressure, and cooling towers reject heat to air.
Modes of heat transfer and heat exchangers	Conduction transfers heat through material; convection involves fluid motion; radiation transfers energy by electromagnetic emission.
Air-standard power cycles	Idealised Carnot, Otto, Diesel and Dual cycles represent energy conversion in heat engines.
IC-engine classification and terminology	Internal-combustion engines can be classified by ignition method, stroke cycle, fuel, cooling, cylinder arrangement and application.
Two-stroke and four-stroke SI/CI engine working	Four-stroke engines separate intake, compression, power and exhaust over two crank revolutions; two-stroke engines complete a cycle more frequently with port/flow-control differences.

LEARNING RESOURCES

References

Recommended thermal-engineering references

1. P. K. Nag, Basic and Applied Thermodynamics.
2. R. K. Rajput, Engineering Thermodynamics.
3. R. S. Khurmi and J. K. Gupta, A Textbook of Thermal Engineering.
4. V. Ganeshan, Internal Combustion Engines.

Approved public thermal-engineering sources

- https://zealpolytechnic.com/wp-content/uploads/2023/04/Thermal_Engineering_I_Full_book_Final.pdf
- https://www.kjei.edu.in/tpp/assets/pdf/ME3K/313310-THERMAL%20ENGINEERING.pdf

These public sources support the explanatory handbook where the official Mechanical Engineering course page was unavailable. They are not a substitute for a restored official Course 3021 detailed syllabus.